AU/PHY/2024/ODD-SEM

Autumn (Odd) Semester Examination -2024 -25

Course Code: PHYUGBS01
Course Title: Engineering Physics
Department of Appearing Students: Engineering, SEM-I, 1st Year
Full Marks: 80 **Time:** 3.00 hrs

Answer any eight questions $10 \times 8 = 80$

1. (a) Define i) unit vector and ii) null vector.
(b) Find a unit vector parallel to the resultant of $\vec{r_1} = 2\hat{\imath} - 4\hat{\jmath} + 5\hat{k}$ and $\vec{r_2} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$
(c) Determine the value of "a" such that $\vec{A} = 2\hat{\imath} + a\hat{\jmath} + \hat{k}$ and $\vec{B} = 4\hat{\imath} - 2\hat{\jmath} - 2\hat{k}$ are mutually perpendicular to each other.
(d) If $\vec{A} = 2\hat{\imath} - 3\hat{\jmath} - \hat{k}$ and $\vec{B} = \hat{\imath} + 4\hat{\jmath} - 2\hat{k}$ find $(\vec{A} + \vec{B}) \times (\vec{A} - \vec{B})$
(2+3+2+3)

2. (a) Define moment of inertia of an object. What is radius of gyration?
(b) State the law of conservation of angular momentum.
(c) Fund the relation between torque and moment of inertia.
(d) State parallel axes theorem of moment of inertia.
(e) Three masses 3 kg, 4 kg and 5 kg are located at the corners of a equilateral triangle of side 1m. Find the center of mass of the system.
((1+1)+1+3+1+3)

3. (a) Define Young's modulus of elasticity. State Hooke's law of elasticity..
(b) State Stoke's law. What is Reynold's number?
(c) State Bernoulli's theorem.
(d) Write down equation of continuity.
(e) The length of a suspended ware increased by 10^{-4} of its original length when a stress of 10^7 Nm^{-2} is applied on it, Calculate the Young's modulus of the material of the wire.
((2+1)+(1+1)+2+1+2)

4. (a) Write down assumptions of the kinetic theory.
(b) Deduce the perfect gas equation.
(c) Find out the relation between rms speeds of the gas molecules with its temperature.
(3+5+2)

5. (a) Write down Zeroth law of thermodynamics.
(b) Write down second law of thermodynamics.
(c) Find out the expression of the efficiency of a Carnot engine.
(d) A Carnot engine takes in heat from a source at a temperature of 100^0 C and rejects heat to a sink at a temperature of 0^0 C. If the engine absorbs 1000 joules from the source, find the work done.
(1+2+4+3)

6. (a) Define Interference of light.
(b) Write down the conditions for observable interference pattern.
(c) Calculate the fringe width using Young's double slit experiment.

OR

In Newton's ring experiment, the diameter of m-th dark ring is 8 mm and the diameter of (m+5)th dark ring is 12 mm. If the radius of curvature of the lower surface of the lens is 10m, find the wavelength of light used.
(2+3+5)

7. (a) Find out the radius of a nucleus whose atomic mass number is 8.
(b) What is Binding energy of a nucleus. Describe how Binding energy of a nucleon depends on the volume and the surface of a nuclei. (With proper mathematical description)
(c) Show that the nuclear mass density is independent of Mass number.
(d) State any two properties of nuclear force.

2+(1+2+2) +2+1

8. (a) Two protons with charge 1.6×10^{-19} c is separated within a nucleus at a distance 1.5×10^{-15}m. Find the electrostatic repulsion force between the protons and the potential energy of the system as well
(b) State the Gauss law of electrostatics and write its mathematical equation
(c) Two particles with charge +2c and +3c separated at a distance 20cm. Another particle with charge 1c is placed at a distance x from the particle with charge +2c on the line of joining where no force is acting on it. Find the value of x

4+3+3

OR

(a) What is a Bravais lattice? How many distinct types of Bravais lattices exist in 3-dimensional space?
(b) Show that the Packing fraction of a Body-centered cubic (BCC) lattices is $\frac{\sqrt{3}\pi}{8}$.
(c) A narrow beam of X-ray with wavelength 1.5 Å is reflected from an ionic crystal with an FCC lattice structure with a density of 3.32 gm/cm^3, The molecular weight is 108 amu. Find the lattice constant?

2+3+5

9. (a) State the Heisenberg Uncertainty Principle.
(b) The uncertainty of detecting the position of a particle is $\Delta x = 10\ nm$. Find the uncertainty of the momentum.
(c) A metal whose work-function(φ) is 4.2 eV is irradiated by radiation whose wavelength is 200 Å. Find the maximum Kinetic Energy of the emitted electrons?
(d) Say, the matter wave of a quantum particle as: $\psi(x,t) = Ae^{i(kx-\omega t)}$. Say the energy operator $\hat{E} = i\hbar\frac{\partial}{\partial t}$ and momentum operator is $\hat{p} = -i\hbar\frac{\partial}{\partial x}$. Using the given operators show that, the energy of the particle is $E = \hbar\omega = h\nu$ and momentum of the particle $p = \hbar k = h/\lambda$

2+2+2+4

10. (a) What is photoelectric effect? Write down the Einstein's equation of Photoelectric effect. Draw the curve between the Stopping Potential (V_s) and the frequency of incident light in the Photo Electric Effect?
(b) When Monochromatic light is shone on a clean metal surface, electrons are emitted from the surface due to the photoelectric effect. State and explain the effect on the emitted electrons of
(i) increasing the frequency of the light.
(ii) increasing the intensity of the light.
(c) A Sodium Vapor lamp emits light photons with a wavelength of $5.89 \times 10^{-7} m$. What is the energy of these photons?

4+4+2

11. (a) State de-Broglie hypothesis. Using the de-Broglie hypothesis, show that the wavelength λ of an alpha particle accelerated through a potential difference V is given by: $\lambda = \frac{h}{\sqrt{16m_p qV}}$, where m_p is the mass of a proton, q is the charge of a proton, and h is the Plank's constant.

(b) What are isobars? How do isotopes of an element differ from one another?

(c) The rate of decay of a radioactive substance is proportional to the number of nuclei present. Using $\frac{dN}{dt} = -\lambda N$, derive the formula $N = N_0 e^{-\lambda t}$, where N_0 is the initial number of nuclei and λ is the decay constant. Using the derived formula calculate the half-life $(t_{1/2})$

3+2+5